

A classical strongly regular Morris Swiss cheese

An affirmative answer to the existential half of Question 1 in arXiv:1412.6406

Autonomous mathematical discovery run `fa_banach_001`
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Status. Candidate full affirmative answer to the first (existential) sentence of Question 1, likely valid, pending expert review. For every $C > 0$ the cheese in Theorem 8.3 can be chosen so that its rational uniform algebra is in fact strongly regular. The universal “must” sentence and the two broader questions in the source remain open.

1 Source question and exact scope

Feinstein, Morley, and Yang prove that for each $C > 0$ there is a classical abstract Swiss cheese $B = ((b_n, s_n))$ with outer disk $\overline{\mathbb{D}}$, with $\mathbb{T} \subset X_B$, with no nonzero bounded point derivations on $R(X_B)$, and with

$$\sum_{n \geq 1} \frac{s_n}{d_n^2} < C, \quad d_n = \text{dist}(B(b_n, s_n), \mathbb{T}). \quad (1)$$

The weighted estimate makes $(f, g) \mapsto \int_{\mathbb{T}} f'(z)g(z) dz$ a bounded bilinear form and hence gives a nonzero derivation into the dual [1]. On source PDF page 18 they ask:

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Combining with (8.3) we conclude that

$$\sum_{n=1}^{\infty} \frac{s_n}{d_n^2} < C.$$

This concludes the proof. □

## 9. Open questions

We raise the following open questions. Let  $X$  be a compact plane set.

QUESTION 1. Let  $B$  be the classical abstract Swiss cheese constructed in Theorem 8.3. Can  $R(X_B)$  be regular? Must  $R(X_B)$  be regular?

QUESTION 2. If  $R(X)$  has no non-zero bounded point derivations, must  $R(X)$  be regular?

Figure 1: Question 1 (and the adjacent Question 2) on page 18 of the locally compiled arXiv source release of arXiv:1412.6406.

**Question 1.** Let  $B$  be the classical abstract Swiss cheese constructed in Theorem 8.3. Can  $R(X_B)$  be regular? Must  $R(X_B)$  be regular?

This packet proves that the answer to *Can it be regular?* is yes, in the stronger form “can be strongly regular.” It does not claim that every choice in Theorem 8.3 is regular.

## 2 Definitions

For a compact plane set  $K$ , let  $R(K)$  be the uniform closure on  $K$  of the rational functions with poles off  $K$ . At  $x \in K$  put

$$M_x = \{f \in R(K) : f(x) = 0\}, \quad J_x = \{f \in R(K) : f \text{ vanishes on a relative neighborhood of } x\}.$$

The algebra is *strongly regular* if  $\overline{J_x} = M_x$  for every  $x$ . For an open disk  $D_j = B(c_j, r_j)$  whose closure is contained in  $\mathbb{D}$ , write

$$d_j = \text{dist}(D_j, \mathbb{T}) = 1 - |c_j| - r_j, \quad W(\{D_j\}) = \sum_j \frac{r_j}{d_j^2}.$$

## 3 Proof intuition

Izzo’s 2022 construction gives two complementary local ideal estimates. A quantitative Körner construction forces  $\overline{J_x} \supset M_x^3$ , while a localized Wermer construction forces  $\overline{M_x^2} = M_x$ . Together they imply strong regularity. Izzo arranged the first estimate with a weighted boundary sum on a square [2]; the first new point here is that a Cayley chart flattens the unit circle *exactly*. Because a Möbius map takes disks to disks, its half-plane version transfers the boundary construction to the unit disk with the same type of weighted control.

The resulting cheese need not be classical. The second new point is a short fixed-boundary classicalisation lemma. During the standard merging process, every aggregate disk has radius at most the sum of the original radii assigned to it. A small value of  $\sum r_j/d_j^2$  then forces that aggregate to remain at least half as far from  $\mathbb{T}$  as its closest assigned disk. Thus the outer unit disk never has to shrink, and the weighted sum increases by at most a factor nine. Strong regularity survives because the two ideal estimates used to obtain it pass to compact subsets.

## 4 A weighted unit-disk form of Izzo’s construction

We isolate the disk version needed below. Its proof records the changes to Izzo’s square argument, rather than treating the change of boundary as a formal affine transformation (which it is not).

**Proposition 1.** *For every  $\varepsilon > 0$  there is a sequence of open disks  $D_j$  with  $\overline{D_j} \subset \mathbb{D}$  such that, for*

$$K = \overline{\mathbb{D}} \setminus \bigcup_j D_j,$$

*one has  $W(\{D_j\}) < \varepsilon$  and, for every  $x \in K$ ,*

$$\overline{J_x(R(K))} \supset M_x(R(K))^3, \quad \overline{M_x(R(K))^2} = M_x(R(K)). \quad (2)$$

*Consequently  $R(K)$  is strongly regular.*

*Proof.* We use the notation and quantitative Körner lemmas of Izzo [2, Lemmas 3.1, 3.5, 4.1, and 6.7]. Those lemmas have two features that matter here. Given a center  $a$ , an error  $\eta > 0$ , and a sufficiently small scale  $\rho$ , they produce pairwise disjoint pole disks in

$$\rho < |w - a| < 2\rho$$

and a function which vanishes on  $|w - a| \leq \rho$  and approximates the relevant coordinate power. In the boundary-weighted version the power is three and

$$\sum_Q \frac{r(Q)}{s_a(Q)^2} < \frac{4\delta}{\rho}, \quad (3)$$

where  $s_a(Q)$  is the distance from  $Q$  to the two coordinate lines through  $a$ . The small parameter  $\delta$  is free.

We first explain the only genuinely new local step, at a point  $a \in \mathbb{T}$ . After rotation use the Cayley map

$$\Phi_a(z) = \frac{a - z}{a + z}.$$

It maps  $a$  to 0,  $-a$  to infinity,  $\mathbb{D}$  to the right half-plane, and  $\mathbb{T}$  to the imaginary axis. Apply the boundary-weighted local construction at 0. No pole disk meets either coordinate axis (otherwise the left side of (3) would be infinite). Retain the disks in the right half-plane and discard those in the left half-plane. Their inverse images under  $\Phi_a$  are disks with closures in  $\mathbb{D}$ ; the discarded poles map outside  $\mathbb{D}$ .

On  $|w| \leq 2\rho$ , the inverse Cayley map and its derivative have uniform upper and lower bounds. It follows directly from the formula for the image of a circle under a Möbius map that, with  $Q^* = \Phi_a^{-1}(Q)$ ,

$$\frac{r(Q^*)}{\text{dist}(Q^*, \mathbb{T})^2} \leq A \frac{r(Q)}{\text{dist}(Q, i\mathbb{R})^2} \leq A \frac{r(Q)}{s_0(Q)^2}, \quad (4)$$

where the absolute constant  $A$  is independent of  $a$  and of small  $\rho$ . If  $h$  is Izzo's local function, set

$$G(z) = (z - a)^3 h(\Phi_a(z)).$$

Since  $z - a = -2aw/(1 + w)$  and  $\text{Re } w \geq 0$  on  $\mathbb{D}$ ,

$$|(z - a)^3 - G(z)| \leq 8|w^3(1 - h(w))| \leq 8\eta.$$

The apparent singularity at  $-a$  is removable: the strengthened Körner function has  $h(w) = 1 + O(w^{-3})$  at infinity. Its rational approximants after composition likewise have all poles either in the retained deleted disks or outside  $\mathbb{D}$ . Hence  $G$  belongs to the appropriate rational algebra, vanishes on a relative neighborhood of  $a$ , and approximates  $(z - a)^3$ .

Cover  $\mathbb{T}$  at the  $n$ th scale by at most  $A_1/\rho_n$  such Cayley neighborhoods. Take  $\eta_n \downarrow 0$ ,  $\rho_n \downarrow 0$ , and then choose the free parameters  $\delta_n$  so fast that

$$\sum_n A A_1 \frac{\delta_n}{\rho_n^2} < \frac{\varepsilon}{4}.$$

Equations (3)–(4) give the desired weighted budget at the boundary, and the preceding approximation gives  $\overline{J}_x \supset M_x^3$  for every  $x \in \mathbb{T}$ . More explicitly, the inverse Cayley image of  $|w| < \rho_n$  contains a Euclidean neighborhood of its center of radius comparable to  $\rho_n$ ; applying Izzo's local criterion [2, Lemma 4.1] converts the covering approximations of  $(z - a)^3$  into the asserted ideal inclusion at every covered point.

For interior points, exhaust  $\mathbb{D}$  by the disks  $L_n = \{|z| \leq 1 - 1/n\}$ . Cover  $L_n$  by at most  $A_2/\sigma_n^2$  disks of radius  $\sigma_n$ , with centers in  $L_n$  and with  $4\sigma_n < 1/n$ , and use Izzo's ordinary local lemma at every covering center. (A maximal separated net gives an absolute  $A_2$ .) Each pole disk then has distance at least  $1/(2n)$  from  $\mathbb{T}$ . The ordinary radius bound therefore makes the total weighted cost at stage  $n$  at most a fixed constant times  $n^2\delta_n/\sigma_n$ . Choosing  $\delta_n$  sufficiently fast gives total cost below  $\varepsilon/4$  and gives

the stronger inclusion  $\overline{J_x} \supset M_x^2$  at every interior point. Together with the boundary construction this proves the first inclusion in (2) everywhere.

For the second inclusion, take a countable cover  $U_j$  of  $\mathbb{D}$  by disks with  $\overline{U_j} \subset \mathbb{D}$ . Apply the scaled Wermer construction in each  $U_j$ , choosing its ordinary radius sum below

$$2^{-j-2}\varepsilon \operatorname{dist}(U_j, \mathbb{T})^2.$$

The total added weighted cost is below  $\varepsilon/4$ . Hallström's local criterion, exactly as in Izzo's proof of his weighted Wermer theorem [2, Theorem 6.6], gives  $\overline{M_x^2} = M_x$  at every interior point. Every boundary point is a peak point, using  $(1 + \bar{a}z)/2$ , so the same equality holds on  $\mathbb{T}$ .

Finally take the union of the two pole families. Izzo's compact-subset lemmas [2, Lemmas 5.1 and 5.2] show that both ideal properties pass to this intersection. The second one implies that  $M_x^3$  is dense in  $M_x$ ; the first then yields  $\overline{J_x} = M_x$  for every  $x$ . Rescaling the three budget allocations slightly, if necessary, makes their total less than the stated  $\varepsilon$ .  $\square$

## 5 Weighted classicalisation without moving the outer circle

The following quantitative observation is the step that prevents ordinary classicalisation from losing  $\mathbb{T}$ .

**Lemma 2.** *Let  $A$  be a unit-disk Swiss cheese whose deleted disks  $D_j = B(c_j, r_j)$  satisfy  $\overline{D_j} \subset \mathbb{D}$  and*

$$\theta = \sum_j \frac{r_j}{d_j^2} < \frac{1}{6}.$$

*Then there is a classical unit-disk Swiss cheese  $B$  such that*

$$\mathbb{T} \subset X_B \subset X_A, \quad \sum_{E \in B} \frac{r(E)}{\operatorname{dist}(E, \mathbb{T})^2} \leq 9\theta.$$

*The outer disk remains exactly  $\overline{\mathbb{D}}$ .*

*Proof.* Run the standard Feinstein–Heath/Mason classicalisation by combining two inner disks whenever their closures meet [1]. Keep the usual allocation bookkeeping: every current inner disk  $E = B(c, R)$  is assigned a disjoint set  $I(E)$  of original disks, contains each assigned disk, and satisfies

$$R \leq \sum_{i \in I(E)} r_i. \tag{5}$$

These invariants are preserved because the minimal disk containing two intersecting disks has radius at most the sum of their radii. They also pass to the limit stages in the standard construction.

Put  $e = \operatorname{dist}(E, \mathbb{T}) = 1 - |c| - R$  and  $d_* = \inf_{i \in I(E)} d_i$ . Containment of  $D_i$  in  $E$  gives

$$e \leq d_i \leq e + 2R \leq d_* + 2R \quad (i \in I(E)).$$

Consequently (5) gives

$$R \leq \theta(d_* + 2R)^2, \quad R \leq \sum_i r_i \leq \theta, \tag{6}$$

because  $0 < d_i < 1$ . If  $R \geq d_*/4$ , then

$$R \leq 36\theta R^2, \quad \text{so} \quad 1 \leq 36\theta R \leq 36\theta^2 < 1,$$

a contradiction. Thus  $R < d_*/4$  and

$$e \geq d_* - 2R > d_*/2 > 0. \quad (7)$$

Every aggregate disk therefore remains strictly inside  $\mathbb{D}$ . The operation that shrinks or moves the outer disk is never invoked, so the classical output has outer disk exactly  $\overline{\mathbb{D}}$  and contains  $\mathbb{T}$ .

For every final disk  $E$ , equations (7) and  $d_i < d_* + 2R < 3d_*/2$  give

$$\frac{R}{e^2} < \frac{4R}{d_*^2} \leq 9 \sum_{i \in I(E)} \frac{r_i}{d_i^2}.$$

The final allocations are disjoint. Summing proves the factor-nine bound. The usual classicalisation inclusion gives  $X_B \subset X_A$ .  $\square$

## 6 Main theorem

**Theorem 3.** *For every  $C > 0$  there is a classical abstract Swiss cheese  $B = ((b_n, s_n))$  such that*

1.  $b_0 = 0$ ,  $s_0 = 1$ , and  $\mathbb{T} \subset X_B$ ;
2.  $R(X_B)$  is strongly regular (hence regular and has no nonzero bounded point derivations);
3. with  $d_n = \text{dist}(B(b_n, s_n), \mathbb{T})$ ,

$$\sum_{n \geq 1} \frac{s_n}{d_n^2} < C;$$

4.  $R(X_B)$  is not weakly amenable.

Thus the existential sentence of Question 1 in *arXiv:1412.6406* has an affirmative answer.

*Proof.* Choose

$$0 < \varepsilon < \min\{1/6, C/9\}$$

and take the unit-disk Swiss cheese  $A$  from Proposition 1. Apply Lemma 2 to obtain a classical  $B$  with  $X_B \subset X_A$ , fixed outer disk, and weighted sum below  $9\varepsilon < C$ .

The two ideal conditions in (2) pass from  $X_A$  to the compact subset  $X_B$  by the same restriction argument used in [2, Lemmas 5.1 and 5.2]. They again imply  $\overline{J_x(R(X_B))} = M_x(R(X_B))$  for all  $x \in X_B$ . Thus  $R(X_B)$  is strongly regular. In particular it is regular and has no nonzero bounded point derivations.

The Cauchy estimate of Feinstein–Morley–Yang [1, Lemma 8.2], together with the weighted sum, gives

$$\left| \int_{\mathbb{T}} f'(z)g(z) dz \right| \leq 4\pi C \|f\|_{X_B} \|g\|_{X_B} \quad (f, g \in R_0(X_B)).$$

There is at least one deleted disk because  $X_B \subset X_A \neq \overline{\mathbb{D}}$ . Choose  $q$  in one of them. For  $f(z) = 1/(z - q)$  and  $g(z) = z - q$  the integral is  $-2\pi i$ . Hence the associated bounded derivation  $R(X_B) \rightarrow R(X_B)^*$  is nonzero, so  $R(X_B)$  is not weakly amenable. All conclusions of Theorem 8.3, with the strengthened regularity conclusion, now hold.  $\square$

## 7 Verification, limitations, and novelty status

The proof has four independent checkpoints.

1. The Cayley map sends the protected circle exactly to an axis and the unit disk to one half-plane; the removability at the antipodal point uses the third-order normalization already present in Izzo’s weighted construction.
2. The interior and boundary covering counts are respectively  $O(\sigma^{-2})$  and  $O(\rho^{-1})$ . The free Körner parameters can be chosen after the scales, so both weighted series converge with an arbitrarily small sum.
3. In Lemma 2, the contradiction is strict because  $\theta < 1/6$ , and the factor nine follows from the two sharp comparisons  $e > d_*/2$  and  $d_i < 3d_*/2$ .
4. Strong regularity is transferred through the two separate inherited ideal statements, not through an unsupported assertion that strong regularity is hereditary for arbitrary uniform algebras.

No computation is used in the proof. The source PDF and the decisive Izzo paper are included in this packet. A bounded novelty search covered the exact wording of Question 1, “classical Swiss cheese strongly regular not weakly amenable,” the source authors, arXiv:1501.04008, arXiv:1503.03785, arXiv:1507.01779, and arXiv:2211.14684, using the local corpus and exact web searches through 11 August 2026. The 2016 paper repeats the broader Question 2; Izzo proves the square weighted theorem but does not state the classical unit-disk conclusion. No prior statement of Theorem 3 was found. This is a bounded search, not a claim of exhaustive novelty.

The result answers only the existential “Can” sentence of source Question 1. It does not show that every cheese produced by Theorem 8.3 is regular, does not answer whether absence of bounded point derivations always implies regularity, and does not answer whether weak amenability forces triviality.

**Human-review recommendation.** Check first the Möbius transfer in Proposition 1, especially removability at  $-a$  and the uniform distance comparison (4). Then check that the standard classicalisation allocation process has exactly the containment and radius invariants used in Lemma 2. If those checks pass, accept the existential half of Question 1 as fully solved.

## References

- [1] J. F. Feinstein, S. Morley, and H. Yang, *Swiss Cheeses and Their Applications*, arXiv:1412.6406, Contemporary Mathematics 645 (2015), DOI: 10.1090/conm/645/12928.
- [2] A. J. Izzo, *A sharper Swiss cheese*, arXiv:2211.14684 (revised 2025).
- [3] J. F. Feinstein, S. Morley, and H. Yang, *Abstract Swiss Cheese Space and Classicalisation of Swiss Cheeses*, J. Math. Anal. Appl. 438 (2016), 119–141; arXiv:1503.03785.